Deliberation beyond shallow consensus

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1 Introduction

Democratic deliberation requires faithfully representing publics' opinions, including where people disagree. The methods currently used to aggregate public input — both consensus protocols and LLM-based summarization — often do not faithfully represent the structure of disagreement they are being asked to mediate. These systems fall prey to preference collapse, which amplifies majority views and flattens minority ones (Shi et al., 2025), and to what Fazelpour and Fleisher (2025) call perspectival homogenization. Most popular forms of pluralistic alignment treat the inconsistency brought by disagreement as a defect to be corrected rather than the source of the exchange that brings democratic synthesis.

Governments are starting to deploy AI in real deliberative processes, using AI tools to help officials extract meaning from large volumes of qualitative citizen input. There is great potential for empowering democracy using these systems, but the technologies are clearly not ready yet to faithfully represent people's opinions in the deliberative process.

This is at odds with how democratic theory values disagreement. Mouffe argues that "the ideal of a pluralist democracy cannot be to reach a rational consensus," and that "agonistic confrontation is the very condition of existence" of democracy itself (Mouffe, 1999). Confrontation is constitutive of democratic life rather than an obstacle to it: according to John Stuart Mill, even a true opinion that never collides with dissent becomes "dead dogma, not a living truth" (Mill, 1859).

We propose a multi-round deliberative framework grounded in Baryshnikov's (1993) unification of Arrow's impossibility theorem (Arrow, 1963) with Chichilnisky's topological social choice (Chichilnisky, 1980). It measures and surfaces disagreement at scale. We develop the theoretical groundwork and a simple implementation built on CIP's Global Dataset (Collective Intelligence Project, 2025) with a latent space pipeline that maps opinions into values. It is at the level of values, rather than of votes, that agreement and disagreement become legible.

Take an example that shows what sets our approach apart: if US, Portuguese, and Brazilian respondents all vote in favor of "AI should be regulated" but their free-form justifications imply different underlying models of what regulation is for (consumer protection, sovereignty, labor), our method will separate the two facts: that the parties agree at the vote level, and how they are inconsistent in the values behind it.

Importantly, our framework does not resolve the disagreement by choosing among the three justifications, because we believe deliberation is of human interest, which renders a fully automated deliberative process impossible. We do believe, however, that the full proposal allows for democracy at scale and solves the democratic trilemma proposed by Fishkin (2009), increasing participation, deliberation and political equality together.

Across several fields of literature, the importance of global inconsistency has been raised as a key issue of study. Rather than merely the kind of direct inconsistency of two contradicting answers, a global inconsistency is more like the paradoxical structure of the Liar paradox—"This sentence is true"—in which the inconsistency is more indirect (Abramsky et al., 2015). In this project we aim to surface such global inconsistency, which we believe is a crucial feature of the kind of obstruction to consensus experienced in everyday deliberation.

2 Methods

This proposal sits within a larger framework that addresses democratic analysis at scale. The full framework consists of five phases:

1. **Framing.** A convening body poses a problem and an initial set of concrete options X_1.
2. **Elicitation (round 1).** A sampled public responds in open text and records positions on X_1.
3. **Analysis.** Analysis. The open-text responses are transformed into dissent graphs, measured for global inconsistency, and embedded in a latent space to produce a legible report of the values that drive the disagreements.
4. **Synthesis.** Human stakeholders read the report and design a revised option set X_2.
5. **Elicitation (round 2).** The same instrument is re-administered over X_2, and the dissent measures are recomputed.

This work focuses on developing the methods for phase 3 (Analysis), while using CIP's Global Dialogues dataset (Collective Intelligence Project, 2025) as a source to phases 1 and 2. We additionally run a small-scale test of phases 4 and 5 over a simulated population.

2.1 Framing & Elicitation

In this work the data for these phases are supplied by CIP's Global Dialogues dataset (Collective Intelligence Project, 2025). In a real deployment, framing and first-round elicitation would be carried out by the convening body and its public.

2.2 Analysis

We set out to take the following steps:

1. Generate or obtain from the global dialogues dataset a set of question contexts. A question context contains a set of questions asked to a participant, or may even indicate a group membership, e.g. demographics. The response statistics are taken per question context rather than per question individually. In general, we expect respondence statistics to a given question to depend on the context that it is in. When this *contextuality* is present, it means that a participant's opinion about a question cannot be considered as a separately exisistant variable, but rather as something negotiated by interaction through the questioning process itself.
2. In the contextuality literature, two types of contextuality are distinguished: true contextuality and direct influence. This may also be understood as global and local inconsistency respectively. We aim to surface such contextuality from our dataset, revealing structural dependencies. This

may be done through one of methods of *contextual fraction* (Abramsky et al., 2017), CbD (Dzhafarov and Kujala, 2016), or PDG inconsistency (Richardson, 2024). Importantly, our measures of contextuality should account for direct influence or local inconsistencies and be able to distinguish them from global inconsistencies. We provide an example of contextuality as follows:

Example A and B both respond that they'd like C, car-free streets. A is positively correlated with C, and so is B, but A and B are anti-correlated with each other, because the underlying reasons why they want car-free streets are opposed: A wants less commercial movement, so that there are fewer people in the streets, while B wants more commercial movement, believing that fewer cars will bring more people.

3. Following the analysis from step 2, which identifies the components relevant to dissent in the graph, we embed the free-form responses into a latent space such as the activation space of an open-source large language model.

4. We examine how the responses are distributed in that space, and how each subgroup's opinions are represented within it. Here, topological data analysis techniques like persistent homology may be applied to look for globally non-trivial structure. A hypothesis is that a structured set of beliefs that are contextually entangled may be associated with non-trivial topological structure. If such structure is present, then we can look for signs of phase transitions in the limit of data.

2.3 Synthesis

In a real deployment, synthesis is carried out by stakeholders helping a population deliberate over a topic. Here we select a small number of the PDGs, analyze their latent space, and write new deliberative prompts that we expect to bridge the dissent between diverse groups. These are then run over the twin system that will do a new deliberation over this data.

2.4 Elicitation & Benchmarking

We compute the delta between the two rounds of deliberation: the change in dissent per topic, the evolution of the discourse, and whether greater levels of synthesis were reached. This serves as a proxy for the quality of the deliberative process and for how much it contributed to agreement across the network.

3 Conclusion

Scaling democracy should not mean scaling the production of consensus, but scaling the legibility of disagreement. This proposal puts forward inconsistency as a foundational element for modeling dissensus.

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